

Second Terminal Examination - 2079

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- Which one of the following set of quantum number is not possible for 4p-electron?
a. $n = 4, l = 1, m = -1, m_s = +\frac{1}{2}$ b. $n = 4, l = 1, m = 0, m_s = +\frac{1}{2}$
c. $n = 4, l = 1, m = 2, m_s = +\frac{1}{2}$ d. $n = 4, l = 1, m = -1, m_s = -\frac{1}{2}$
- If an atom has four unpaired electrons, what is likely to be the total spin of the electron.
a. 1 b. 2.5 c. 2 d. 4
- The oxidation state of Cr in $\text{Cr}(\text{CO})_6$ is
a. 0 b. 2 c. 3 d. 6
- If M-shell has 5 electrons, what would be the name of element.
a. Boron b. Phosphorus c. Sulphur d. Nitrogen
- 'H' loses electron to form H^+ . In this aspect H behaves as.
a. Alkali metal b. Noble gas
c. Alkaline earth metal d. halogens
- Ozone can be detected by using
a. Na b. Hg c. Ag d. Ar
- Which gas of nitrogen is used as an anaesthetic in dental surgery?
a. N_2O b. NO c. NO_2 d. N_2O_5

8. Which one of the hydric acids doesn't form any precipitate with AgNO_3 ?
a. HF b. HCl c. HBr d. HI
9. Which of the following is an example of Heterocyclic compound.
a. benzene b. cyclopentane c. pyrrole d. Phenol
10. The compound having octane number 100 is
a. n-heptane b. Iso-butane
c. 2,2,4-trimethylpentane d. both b & c
11. IUPAC name of the given compound is
 $\text{CH}_3 - \text{O} - \text{CH}_2 - \text{CH}_3$
a. Methoxyethane b. Ethoxymethane
c. Methoxymethane d. Ethoxyethane

Group - B

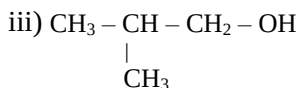
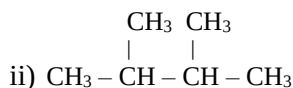
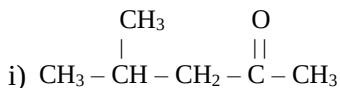
Give short answer to the following questions.

[8 × 5 = 40]

1. a. Oxidation and reduction is the chemical process which is based on transfer of electron.
i. What is disproportionation reaction? Discuss with suitable example. [2]
ii. Differentiate between oxidant and reductant. [1]
b. Arrange the following orbital in ascending order of energy. Discuss with suitable reason. [2]
4f, 3p, 5d, 6s
2. Balance the given redox reaction with the help of the oxidation number method or ion-electron method. Also, find the oxidant and reductant in the given reaction. [4+1]
 $\text{Zn} + \text{NaOH} + \text{NaNO}_3 \longrightarrow \text{Na}_2\text{ZnO}_2 + \text{NH}_3 + \text{H}_2\text{O}$
3. Redox reaction is the chemical reaction in which oxidation and reduction occur simultaneously.
a. Determine the oxidation number of the underlined elements. [2]
i) $(\underline{\text{C}}\text{OOH})_2$ ii) $\underline{\text{S}}\text{O}_4^{--}$
b. Why H_2S acts only as reducing agent whereas SO_2 acts both as oxidizing agent as well as reducing agent? [2]

- c. Split the equation into oxidation half and reduction half with proper reason. [1]
$$\text{Cr}_2\text{O}_7^{--} + \text{Fe}^{++} \longrightarrow \text{Fe}^{+++} + \text{Cr}^{3+}$$
4. Halogens are extremely reactive non-metals and form haloacid with hydrogen.
 - a. Give the balanced chemical equation for the preparation of chlorine in the laboratory. How excess chlorine react with ammonia? [2]
 - b. Write a chemical reaction to show HBr is powerful reducing agent than HCl. [1]
 - c. Write the reactions involved in the test of Cl^- ion. Mention two uses of hydrochloric acid. [2]
5. Hydrogen and oxygen are the abundant elements in the earth.
 - a. Differentiate between Atomic and Nascent hydrogen. [1]
 - b. Classify the following oxide with a suitable reason. [2]
 Na_2O PbO
 - c. Write the resonating structure of ozone. [2]
6. Ammonia is the most valuable compound of nitrogen.
 - a. How ammonia reacts with copper sulphate solution? [2]
 - b. Why is conc. HNO_3 stored in a dark bottle? [1]
 - c. Why does ammonia turn mercurous nitrate paper black? [1]
 - d. Write any two applications of ammonia. [1]
7. Periodic table is used to study different periodic properties.
 - a. Distinguish between electro negativity and electron affinity. [1]
 - b. Which of the following is the smallest in size?
 N^{3-} , Na^+ . Give an appropriate reason. [2]
 - c. Explain: [2]
 - i) Ionization energy decrease in a group from top to bottom.
 - ii) Ionization energy of noble gases are very high.
8. International Union of Pure and Applied Chemistry (IUPAC) developed a system of giving the name to an organic compound called IUPAC system.
 - a. Write the structure of the following IUPAC names, [2]
 - i) 2-methylpropan-2-ol
 - ii) 2-bromopropene

b. Write the IUPAC names of the following compounds, [3]



Group - C

Give long answer to the following questions. [3 × 8 = 24]

9. In the nineteenth century, the atom was regarded as a small indivisible particle, the unit of matter in all chemical changes.
- Briefly, explain the Rutherford nuclear model of an atom. [4]
 - State the “Pauli exclusion principle”. Write all the four-quantum numbers of valence electrons of iron. [1+1]
 - Draw a labeled diagram to show different spectral series of hydrogen spectra. [2]
10. Nitrogen is important element which can form different compounds.
- Explain the principle for the manufacture of nitric acid by the Ostwald’s process. [3]
 - What happens when concentrated HNO_3 reacts with [2]
 - H_2S
 - Zn
 - Draw a flow sheet diagram of the manufacture of ammonia by Haber’s process. [2]
 - Why can’t ammonia be dried using conc. H_2SO_4 ? [1]
11. Organic chemistry is the branch of chemistry which deals with study of hydrocarbons and their derivatives.
- What is Homologous series? Mention the characteristics of homologous series. [3]
 - Write the series containing -OH functional group upto four members with IUPAC name. [2]
 - Define Cetane number. Write two examples of gasoline additives. [2]
 - What is alkyl group? Give example of it. [1]

❧ - The End - ❧